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Air-conditioning outlet device

Abstract

The air-conditioning outlet device includes the housing having a frame shape and having a flow path through which air flows. The housing has an overlapping surface that a fin is overlappable in a state where the flow path is closed, and a clearance recess recessed from the overlapping surface so as not to come into contact with the fin when the fin rotates from the state where the flow path is closed. The clearance recess of the housing and the fin define a blow-in space into which the air is blown from a gap between the housing and the fin. One of the fin and the housing includes a barrier wall that interferes with the air flowing through the blow-in space.

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Background/Summary

FIELD OF THE INVENTION

(1) The present invention relates to an air-conditioning outlet device in which air blown out from an outlet can be shut off by a rotating fin.

BACKGROUND

(2) A vehicle such as a passenger car is provided with an air-conditioning outlet device that is an outlet capable of blowing air out toward a vehicle cabin. A technique relating to the air-conditioning outlet device in the related art has been disclosed.

(3) An air-conditioning outlet device disclosed in Japanese Unexamined Patent Publication No. 2015-152202 includes a housing having a frame shape and having a flow path through air flows; a fin which is disposed in the flow path and by which the flow path is openable and closable; and an operation knob with which the fin can be operated from the outside. The fin includes a plurality of

- plate-shaped portions and shaft portions that protrude from the respective plate-shaped portions in opposite directions and that are supported by the housing.
- (4) When an occupant operates the operation knob from the outside, the fin can rotate (turn) around the shaft portions supported by the housing, to open and close the flow path. When the flow path is closed, the blowing of the air from an outlet to the vehicle cabin can be shut off.
- (5) The fin is rotatably set inside the housing. Further, the fin is attached by fitting the shaft portions of the fin to the housing having a frame shape. For this reason, a slight gap is generated around the shaft portions of the fin. Even in the state where the flow path is closed, the air that has leaked from the gap is blown into the vehicle cabin.
- (6) Particularly, when the housing has a frame shape, namely, when support portions that support the respective shaft portions of the fin are not separated but integrated, in order to allow the fin to be fitted to the housing, the gap around the shaft portions of the fin inevitably becomes large.
- (7) An object of the invention is to provide a technique of being able to suppress the leakage of air from a gap around a shaft portion of a fin in an air-conditioning outlet device including a housing having a frame shape.

SUMMARY

- (8) According to a first aspect, there is provided an air-conditioning outlet device including: a housing having a frame shape and having a flow path through which air flows; and a fin which is disposed in the flow path and by which the flow path is openable and closable. The fin includes a plate-shaped portion and a pair of shaft portions protruding from the plate-shaped portion in opposite directions. The housing includes a pair of support portions that support the respective shaft portions and that are integrated with each other. Each of the support portions includes a base portion having a support hole that rotatably supports the shaft portion, with respect to the base portion, a stepped portion stepped in a direction toward a center of the housing with an axis direction of the shaft portion as a reference, and a connecting portion that connects the base portion and the stepped portion. The connecting portion has an overlapping surface that the plate-shaped portion of the fin is overlappable in a state where the flow path is closed, and a clearance recess recessed from the overlapping surface so as not to come into contact with the fin when the fin rotates from the state where the flow path is closed. In the state where the flow path is closed, the clearance recess of the connecting portion and the plate-shaped portion define a blow-in space into which the air is blown from a gap between the plate-shaped portion and the base portion. One of the fin and the housing includes a barrier wall that interferes with the air flowing through the blow-in space.
- (9) According to a second aspect, in the air-conditioning outlet device according to the first aspect, it is preferable that the shaft portion includes a large-diameter portion having an annular end surface that is contactable with an edge of the support hole, that the barrier wall is formed of a bulging portion that bulges from the fin and the large-diameter portion toward a wall surface of the clearance recess, and that the bulging portion has a facing surface facing the wall surface of the clearance recess.
- (10) According to a third aspect, in the air-conditioning outlet device according to the second aspect, it is preferable that a lightening recess is formed on an opposite side of the fin from the bulging portion.
- (11) According to a fourth aspect, in the air-conditioning outlet device according to the first aspect, it is preferable that the barrier wall is formed of a fin-side lid portion provided on the plate-shaped portion so as to close the blow-in space.
- (12) According to a fifth aspect, in the air-conditioning outlet device according to the first aspect, it is preferable that the barrier wall is formed of a housing-side lid portion provided on the housing so as to close the blow-in space.
- (13) The air-conditioning outlet device includes the housing having a frame shape and having the flow path through which the air flows. The housing includes the pair of support portions that

support the respective shaft portions and that are integrated with each other. Each of the support portions includes the base portion having the support hole that rotatably supports the shaft portion, the stepped portion stepped with respect to the base portion, and the connecting portion that connects the base portion and the stepped portion.

(14) The connecting portion has the overlapping surface that the plate-shaped portion of the fin is overlappable in the state where the flow path is closed, and the clearance recess recessed from the overlapping surface so as not to come into contact with the fin when the fin rotates from the state where the flow path is closed.

(15) In the state where the flow path is closed, the clearance recess of the connecting portion and the plate-shaped portion define the blow-in space into which the air is blown from the gap between the plate-shaped portion and the base portion. One of the fin and the housing includes the barrier wall that interferes with the air flowing through the blow-in space.

(16) With the above configuration, the flow of the air that has flowed from a gap between the fin and the housing into the blow-in space is interfered with by the barrier wall. The leakage of the air around the shaft portion can be suppressed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a perspective view of an air-conditioning outlet device according to a first embodiment;

(2) FIG. 2 is an exploded perspective view of the air-conditioning outlet device illustrated in FIG. 1;

(3) FIG. 3 is a view of a housing and fins as viewed from an upstream side;

(4) FIG. 4 is a view of the housing to which the fins are attached, as viewed from the upstream side;

(5) FIG. 5 is a cross-sectional view taken along line 5-5 in FIG. 4;

(6) FIG. 6 is a cross-sectional view taken along line 6-6 in FIG. 5;

(7) FIG. 7 is a view for describing a configuration of a shaft portion of the fin and of the periphery of a support hole of a base portion of the housing;

(8) FIG. 8 is a cross-sectional view taken along line 8-8 in FIG. 4;

(9) FIG. 9A is an enlarged view of a portion surrounded by line 9A in FIG. 4; FIG. 9B is a view for describing a blow-in space; FIG. 9C is a cross-sectional view taken along line 9C-9C in FIG. 9B;

(10) FIG. 10A is a view for describing an air-conditioning outlet device according to a second embodiment; FIG. 10B is a view for describing an action of the air-conditioning outlet device according to the second embodiment;

(11) FIG. 11A is a view for describing the air-conditioning outlet device according to the third embodiment; and FIG. 11B is a view for describing an action of the air-conditioning outlet device according to the third embodiment.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

(12) Embodiments of the invention will be described with reference to the accompanying drawings. Incidentally, in the drawings, Fr represents the front, Rr represents the rear, L represents the left, R represents the right, Up represents the top or up, and Dn represents the bottom or down. Further, with the flow direction of air inside a housing as a reference, an upstream side is referred to as the rear and a downstream side is referred to as the front.

First Embodiment

(13) FIG. 1 illustrates an air-conditioning outlet device **10** disposed in front of an occupant in a passenger car. The air-conditioning outlet device **10** is a resin-molded product, and can shut off air blown out from an outlet to a vehicle cabin, when an operation knob **14** to be described later is slid

leftward and rightward

(14) Referring to FIG. 2, the air-conditioning outlet device **10** includes a housing **20** having a frame shape and having a flow path **21** through which air flows; a fin unit **19** which is disposed in the flow path **21** and by which the flow path **21** is openable and closable; an outlet-side member **11** that can be attached to an outlet **24** of the housing **20**; and the operation knob **14** which can be attached to the outlet-side member **11** and with which the occupant can operate the fin unit **19** from the outside.

(15) The outlet-side member **11** includes a body portion **12** having a frame shape, and an elongated portion **13** having a plate shape and provided to partition the inside of the body portion **12** in a longitudinal direction.

(16) The operation knob **14** includes a pair of knob halves **15** and **15** that sandwich the elongated portion **13** therebetween, and a connector **16** that is supported by the pair of knob halves **15** and **15** and that connects the operation knob **14** and the fin unit **19**.

(17) The fin unit **19** includes a first fin **31** to a fifth fin **35** each having a plate shape, and a link **36** that connects the first fin **31** to the fifth fin **35** to each other to synchronize the directions of each of the fins. Incidentally, the number of the fins can be changed as appropriate.

(18) Referring to FIG. 3, the first fin **31** to the fifth fin **35** have the same basic configuration.

Hereinafter, the first fin **31** to the fifth fin **35** will be described as a “fin **30**”. The fin **30** is configured such that a plate-shaped portion **60** and a pair of shaft portions **70** and **70** protruding from the plate-shaped portion **60** in opposite directions are integrated.

(19) The housing **20** is configured such that a pair of support portions **40** and **40** which extend parallel to each other and which can support the respective fins **30** and side walls **49** and **49** which extend orthogonal to the support portions **40** and **40** and which connect both ends of the support portions **40** and **40** are integrated.

(20) Referring to FIG. 4 and FIG. 5, when the operation knob **14** is moved leftward and rightward, the connector **16** fixed to the operation knob **14** also moves leftward and rightward. A groove **17** of the connector **16** sandwiches a round rod-shaped portion **33a** of a third fin **33**. When the round rod-shaped portion **33a** moves along the groove **17**, the third fin **33** rotates around the shaft portions **70** as the center. The first fin **31**, the second fin **32**, the fourth fin **34**, and the fifth fin **35** connected to the third fin **33** via the link **36** swing around the respective shaft portions **70** as the centers.

(21) Referring to FIG. 5, with the flow direction of air as a reference (refer to arrow **(1)** in FIG. 5), in a state where the flow path **21** is closed, an upstream side of the shaft portions **70** of the respective fins **30** (**31** to **35**) is referred to as an upstream flow path **22**, and a downstream side of the shaft portions **70** of the respective fins **30** (**31** to **35**) is referred to as a downstream flow path **23**.

(22) Referring to FIG. 3 and FIG. 5, the plate-shaped portion **60** is configured such that a center portion **61** between the shaft portions **70** and **70**, an upstream fin **62** located outside the center portion **61** in a radial direction of the shaft portions **70** to swing in the upstream flow path **22**, and a downstream fin **65** located outside the center portion **61** in the radial direction of the shaft portions **70** to swing in the downstream flow path **23** are integrated.

(23) Referring to FIG. 3 and FIG. 6, each of the support portions **40** is configured such that a base portion **41** having support holes **42** capable of supporting the shaft portion **70**, a stepped portion **47** stepped with respect to the base portion **41**, and a connecting portion **50** which connects the base portion **41** and the stepped portion **47** are integrated. With a direction in which an axis AX of the shaft portions **70** extends (up-down direction) as a reference, the stepped portion **47** is closer to a center CL of the housing **20** than the base portion **41**. The base portion **41** has guide grooves **45** that guide the shaft portions **70** to the support holes **42** when the fin unit **19** is fitted and attached to the housing **20**.

(24) Referring to FIG. 7, the connecting portion **50** of the support portion **40** has an overlapping surface **51** that overlaps a front surface **63** of the upstream fin **62** in a state where the flow path **21**

is closed (refer to FIG. 5), and a clearance recess 52 recessed from the overlapping surface 51.

(25) The clearance recess 52 is open toward the upstream side (the rear), and is set such that the support portion 40 does not come into contact with the center portion 61 or the shaft portions 70 of the fin 30 when the fin 30 rotates. A wall surface 53 of the clearance recess 52 is an arc-shaped surface centered on the axis AX.

(26) Referring to FIG. 7 and FIG. 8, the shaft portion 70 is configured such that a small-diameter portion 71 of which an outer peripheral surface 72 is rotatably supported on an inner peripheral surface 43 of the support hole 42 and a large-diameter portion 73 which has a larger diameter than that of the small-diameter portion 71 and which is located closer to a center CL side of the housing 20 than the small-diameter portion 71 are integrated. An annular outer end surface 74 of the large-diameter portion 73 can come into contact with an edge 44 of the support hole 42. The large-diameter portion 73 regulates the position (axis AX direction) of the fin 30.

(27) Referring to FIG. 9B and FIG. 9C, the wall surface 53 of the clearance recess 52 includes an outer peripheral portion 54 located on an outer peripheral side of an outer peripheral surface 75 of the large-diameter portion 73 of the fin 30, and a non-contact portion 55 that is located closer to the center CL side than the outer peripheral portion 54 and that does not come into contact with the outer peripheral surface 75 of the large-diameter portion 73. In a state where the flow path 21 is closed, a front surface 66 of a center portion 61 of the fin 30, the non-contact portion 55 of the clearance recess 52, and an inner end surface (surface facing the center CL) of the large-diameter portion 73 define a blow-in space S having a semicircular columnar shape. Incidentally, for convenience of description of the blow-in space S, a “barrier wall 80” to be described later is not drawn in FIG. 9B and FIG. 9C.

(28) Referring to FIG. 9A, the fin 30 is rotatably provided inside the housing 20. For this reason, a slight gap is generated between an end surface 64 (refer to also FIG. 7) of the upstream fin 62 of the fifth fin 35 and an inner surface 46 (surface facing the center CL side) of the base portion 41. As indicated by arrow (2), air can enter the gap.

(29) Referring to FIG. 9B and FIG. 9C, a gap is also generated at a location where the boundary between the large-diameter portion 73 and the end surface 64 of the upstream fin 62 and the boundary between the clearance recess 52 and the overlapping surface 51 overlap each other. The air that has flown between the end surface 64 and the inner surface 46 uses the gap as an inlet E, and the air is blown into the blow-in space S.

(30) Referring to FIG. 7 and FIG. 8, the fin 30 includes the barrier wall 80 that interferes with the air flowing from the blow-in space S (refer to FIG. 9B) toward the downstream flow path 23. The barrier wall 80 is formed of a bulging portion 81 that bulges from the center portion 61 of the fin 30 and the large-diameter portion 73 toward the wall surface 53 of the clearance recess 52. In other words, the bulging portion 81 narrows an air flow path by filling the blow-in space S.

(31) The bulging portion 81 has a facing surface 82 facing the wall surface 53 of the clearance recess 52. In other words, it can be said that the facing surface 82 of the bulging portion 81 is a surface extending from an edge 75a of the outer peripheral surface 75 of the large-diameter portion 73 to the center CL side of the housing 20. The facing surface 82 is an arc-shaped surface centered on the axis AX, and is a surface along the wall surface 53 of the clearance recess 52. It is preferable that a slight gap D which does not interfere with the rotation of the fin 30 is provided between the facing surface 82 of the bulging portion 81 and the wall surface 53 of the clearance recess 52. An upper surface 83 of the bulging portion 81 is located on substantially the same plane as an inner surface 48 (surface facing the center CL side) of the stepped portion 47.

(32) As described above, the gap D is set between the facing surface 82 of the bulging portion 81 and the wall surface 53 of the clearance recess 52. Even when the air is blown into the blow-in space S (refer to FIG. 9B and FIG. 9C) from the inlet E, as indicated by arrow (3), the blown-in air flows through the gap D between the facing surface 82 of the bulging portion 81 and the wall surface 53 of the clearance recess 52. Compared to the case where the bulging portion 81 (barrier

wall **80**) is not provided, the flow speed of the air also becomes slower, and the leakage of the air around the shaft portions **70** can be suppressed.

(33) Referring to FIG. **7** and FIG. **8**, the center portion **61** of the fin **30** includes a lightening recess **84** recessed from a rear surface **67** of the center portion **61** of the fin **30** toward the bulging portion **81**. Namely, the lightening recess **84** is formed on a side opposite the bulging portion **81**. The bulging portion **81** has a box shape that is open toward the upstream flow path **22**. The time of a step of forming the fin **30** (**31** to **35**) can be shortened and deformation can also be suppressed.

Second Embodiment

(34) Referring to FIG. **10A**, an air-conditioning outlet device **10A** according to a second embodiment includes a fin **30A**. Configurations that are common to the air-conditioning outlet device **10** of the first embodiment are denoted by the same reference signs as in the first embodiment, and descriptions thereof will not be repeated.

(35) Referring to FIG. **10A** and FIG. **10B**, a barrier wall **80A** is formed of a fin-side lid portion **86** that is provided on the center portion **61** of the fin **30A** to close the blow-in space **S**. The fin-side lid portion **86** has a semicircular plate shape, and the thickness and the diameter of the fin-side lid portion **86** are substantially the same as those of the large-diameter portion **73**. An upper surface **87** (surface facing the center CL side) of the fin-side lid portion **86** is located on substantially the same plane as (namely, is flush with) the inner surface **48** of the stepped portion **47**. It is preferable that the slight gap **D** which does not interfere with the rotation of the fin **30A** is provided between an outer peripheral surface **86a** of the fin-side lid portion **86** and the wall surface **53** of the clearance recess **52**.

(36) As indicated by arrow **(3)**, the flow of the air that has flowed from the inlet **E** into the blow-in space **S** is shut off by the fin-side lid portion **86**. The leakage of the air around the shaft portions **70** can be suppressed.

Third Embodiment

(37) Referring to FIG. **11A**, an air-conditioning outlet device **10B** according to a third embodiment includes a fin **30B** and a housing **20B**. Configurations that are common to the air-conditioning outlet device of the first embodiment are denoted by the same reference signs as in the first embodiment, and descriptions thereof will not be repeated.

(38) Referring to FIG. **11A** and FIG. **11B**, a barrier wall **80B** is formed of a housing-side lid portion **88** that is provided on the housing **20B** to close the blow-in space **S**. The housing-side lid portion **88** is integrated with the wall surface **53** of the clearance recess **52**, and an upper surface **89** (surface facing the center CL side) of the housing-side lid portion **88** is located on substantially the same plane as (namely, is flush with) the inner surface **48** of the stepped portion **47**. A rear surface **89a** (surface facing the upstream side) of the housing-side lid portion **88** is located on (namely, is flush with) substantially the same plane as the overlapping surface **51**.

(39) A clearance groove **68** for avoiding contact with the housing-side lid portion **88** is formed in the fin **30B**. The clearance groove **68** extends inward from an end surface of the center portion **61** of the fin **30B** in the radial direction.

(40) As indicated by arrow **(4)**, the flow of the air that has flowed from the inlet **E** into the blow-in space **S** is shut off by the housing-side lid portion **88**. The leakage of the air around the shaft portions **70** can be suppressed.

(41) Incidentally, the invention is not limited to the embodiments as long as the actions and the effects of the invention are exhibited.

Claims

1. An air-conditioning outlet device comprising: a housing having a frame shape and having a flow path through which air flows; and a fin which is disposed in the flow path and by which the flow path is openable and closable, wherein the fin includes a plate-shaped portion and a pair of shaft

portions protruding from the plate-shaped portion in opposite directions, the housing includes a pair of support portions that support the respective shaft portions and that are integrated with each other, each of the support portions includes a base portion having a support hole that rotatably supports the shaft portion, with respect to the base portion, a stepped portion stepped in a direction toward a center of the housing with an axis direction of the shaft portion as a reference, and a connecting portion that connects the base portion and the stepped portion, the connecting portion has an overlapping surface that the plate-shaped portion of the fin is overlappable in a state where the flow path is closed, and a clearance recess recessed from the overlapping surface so as not to come into contact with the fin when the fin rotates from the state where the flow path is closed, in the state where the flow path is closed, the clearance recess of the connecting portion and the plate-shaped portion define a blow-in space into which the air is blown from a gap between the plate-shaped portion and the base portion, and one of the fin and the housing includes a barrier wall that interferes with the air flowing through the blow-in space.

2. The air-conditioning outlet device according to claim 1, wherein the shaft portion includes a large-diameter portion having an annular end surface that is contactable with an edge of the support hole, the barrier wall is formed of a bulging portion that bulges from the fin and the large-diameter portion toward a wall surface of the clearance recess, and the bulging portion has a facing surface facing the wall surface of the clearance recess.

3. The air-conditioning outlet device according to claim 2, wherein a lightening recess is formed on an opposite side of the fin from the bulging portion.

4. The air-conditioning outlet device according to claim 1, wherein the barrier wall is formed of a fin-side lid portion provided on the plate-shaped portion so as to close the blow-in space.

5. The air-conditioning outlet device according to claim 1, wherein the barrier wall is formed of a housing-side lid portion provided on the housing so as to close the blow-in space.
